PRIME PALINDROME – JAVA PROGRAM

1. Objective

To develop a Java program that accepts an integer from the user, checks whether the number is a palindrome and whether it is prime, and displays whether the given number is a Prime Palindrome.

2. Task Description

Write a Java program using Scanner, loops, and conditional statements to determine whether a given number is both prime and palindrome. The program reverses the number to check the palindrome property and counts the number of divisors to check the prime property.

3. Step-wise Execution

1. Start the program.

2. Create a Scanner object to read input from the keyboard.

3. Ask the user to enter a number and store it in n.

4. Copy n into temp and initialize reverse to 0.

5. Use a while loop to extract each digit using temp % 10 and construct the reversed number.

6. Initialize count to 0.

7. Use a for loop from 1 to n and count how many numbers divide n exactly.

8. If reverse is equal to n and count is equal to 2, the number is both palindrome and prime.

9. Display the appropriate result.

10. Close the Scanner and end the program.

4. Algorithm

Step 1: Start.

Step 2: Read the integer n.

Step 3: Set temp = n and reverse = 0.

Step 4: Repeat while temp > 0: digit = temp % 10, reverse = reverse * 10 + digit, temp = temp / 10.

Step 5: Set count = 0.

Step 6: For i = 1 to n, if n % i == 0, increment count.

Step 7: If reverse == n and count == 2, print Prime Palindrome.

Step 8: Otherwise, print not a Prime Palindrome.

Step 9: Stop.

5. UML Diagram

PrimePalindrome	
+ main(args: String[]): void	
Scanner sc	Reads input from user

n, temp	Input and temporary values
reverse	Stores reversed number
count	Counts divisors

Flow: Start → Read n → Reverse the number → Count divisors → Check (reverse == n AND count == 2) → Display result → Stop

6. Program Code

```
package core_java;

import java.util.Scanner;

public class PrimePalindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int temp = n;
        int reverse = 0;

        while (temp > 0) {
            int digit = temp % 10;
            reverse = reverse * 10 + digit;
            temp = temp / 10;
        }
        int count = 0;

        for (int i = 1; i <= n; i++) {
            if (n % i == 0) {
                count++;
            }
        }

        if (reverse == n && count == 2)
            System.out.println(n + " is a Prime Palindrome");
        else
            System.out.println(n + " is not a Prime Palindrome");

        sc.close();
    }
}
```

7. Output Screen

Example 1:

```
Enter a number: 131
131 is a Prime Palindrome
```

Example 2:

```
Enter a number: 121
121 is not a Prime Palindrome
```

8. Result

The Java program successfully checks whether the given number is both prime and palindrome and displays the corresponding result.

9. Conclusion

This program demonstrates the use of Scanner, while loop, for loop, modulus operator, conditional statements, and basic number logic in Java.